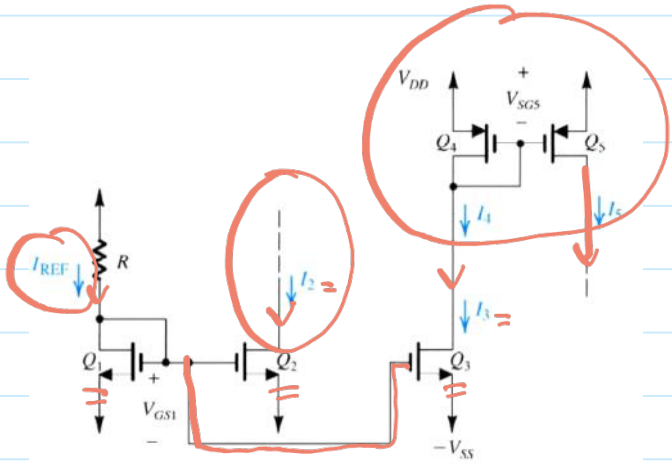
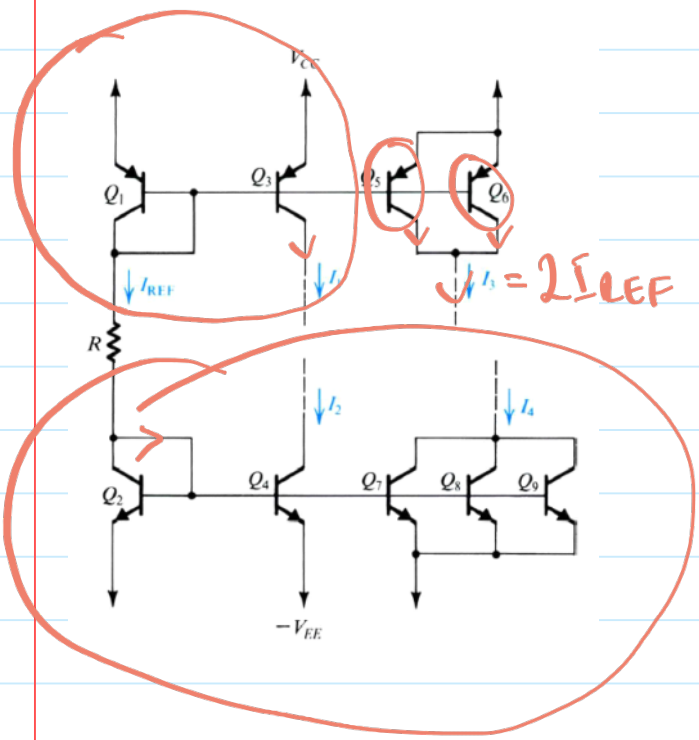
Reading

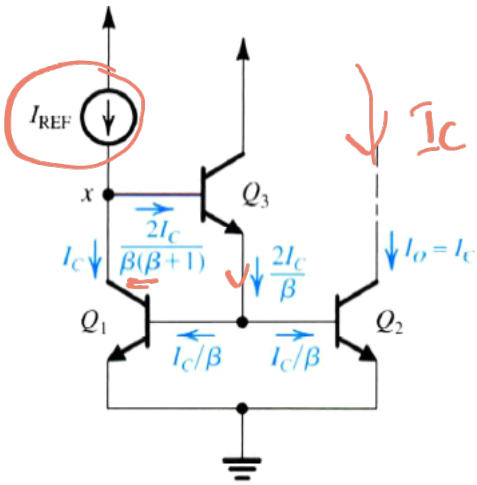
Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 8.3 The Basic Gain Cell
- Chapter 8.4 The Common-Gate and Common-Base Amplifiers as Current Buffers

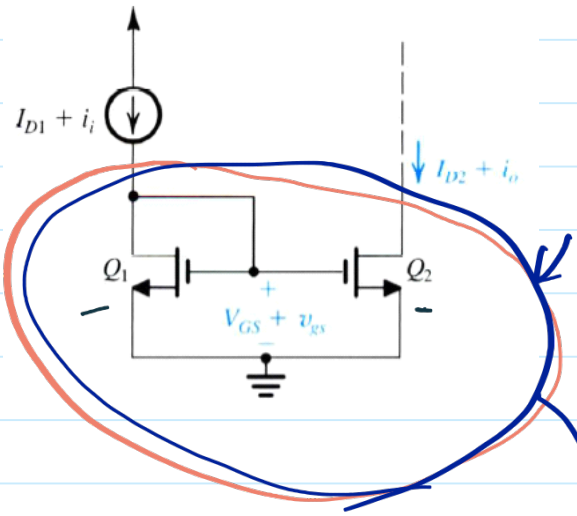
Recap


V_{DD}
V_{SG5}
Q_4
Q_5
I_{REF}
R
I_1
I_5
I_2
I_3
Q_1
Q_2
Q_3
V_{GS1}
$-V_{SS}$


V_{CC}
Q_1
Q_3
Q_5
Q_6
I_{REF}
$I_3 = 2I_{REF}$
R
I_2
I_4
Q_2
Q_4
Q_7
Q_8
Q_9
$-V_{EE}$


I_{REF}
x
Q_3
$\frac{2I_C}{\beta(\beta+1)}$
$\frac{2I_C}{\beta}$
I_C
$I_O = I_C$
Q_1
Q_2
I_C/β
I_C/β

Recap



$$R_{in} = r_{o1} \| \frac{1}{g_{m1}} \simeq \frac{1}{g_{m1}}$$

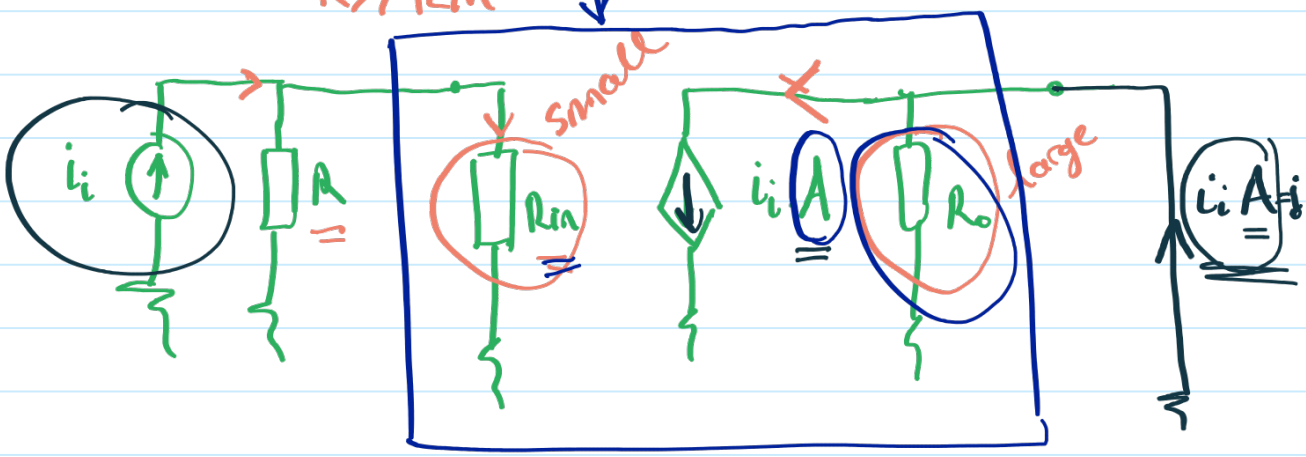
$$R_o \simeq r_{o2}$$

$$A_{is} = \frac{(W/L)_2}{(W/L)_1}\left[\frac{1}{1+(1/g_{m1}r_{o1})}\right]$$

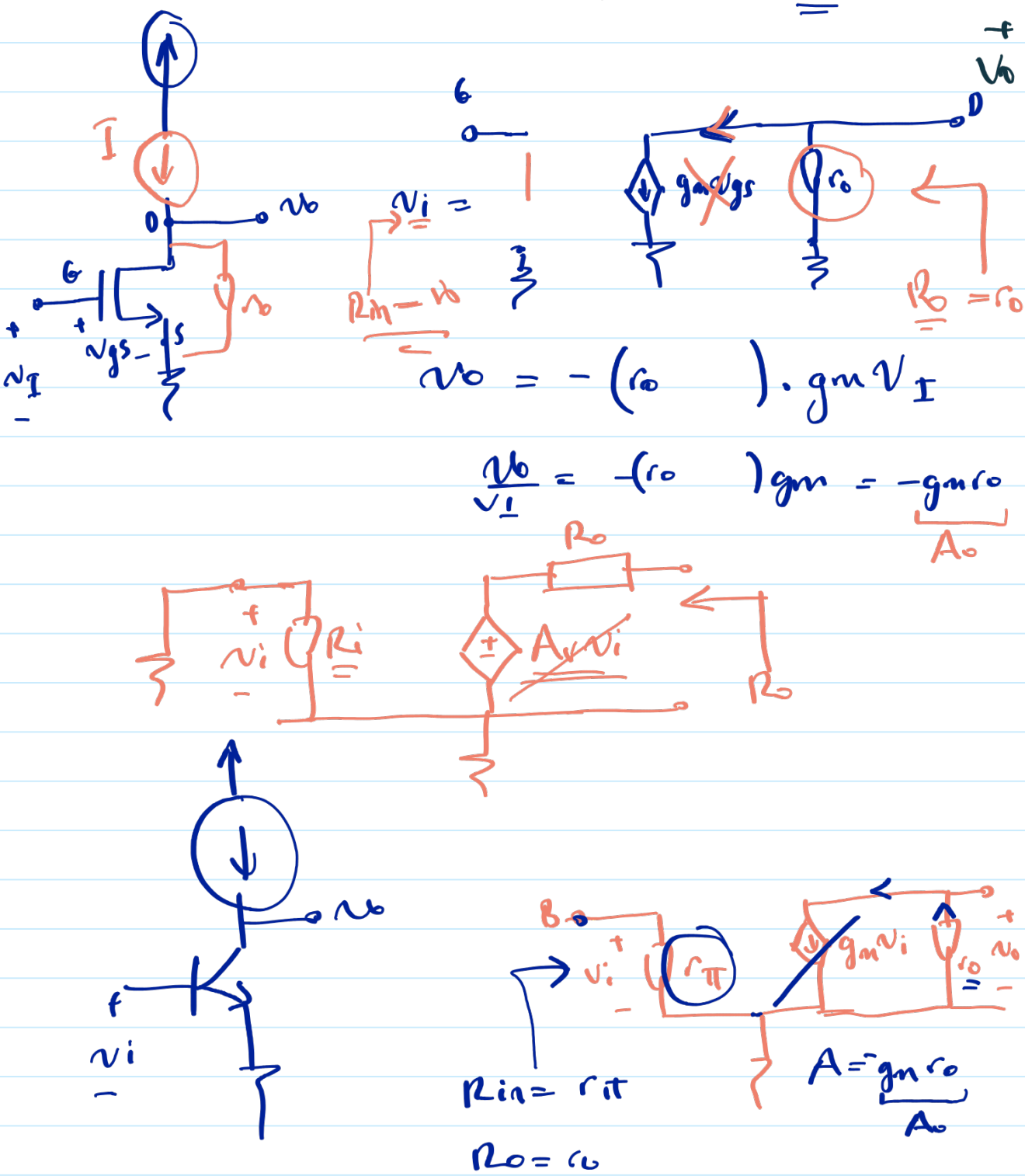
$\frac{i_o}{i_i} = A$

$R_S \gg R_{in}$

$R_o \gg R_L$



CS and CE with Current-Source Load



The Intrinsic Gain

$$A_0 = g_m r_o$$

$$g_m = \frac{I_C}{V_T} \quad V_T \approx 0.025\,V \qquad r_o = \frac{V_A}{I_C}$$

$$A_0 = \frac{I_C}{V_T} \cdot \frac{V_A}{I_C} = \frac{V_A}{V_T} = 200 - 2000$$

BST

MOS

$$A_0 = g_m r_o$$

$$g_m = \frac{I_D}{V_{ov}/2} = \sqrt{2\mu C_{ox} W/L}\ \sqrt{I_D}$$

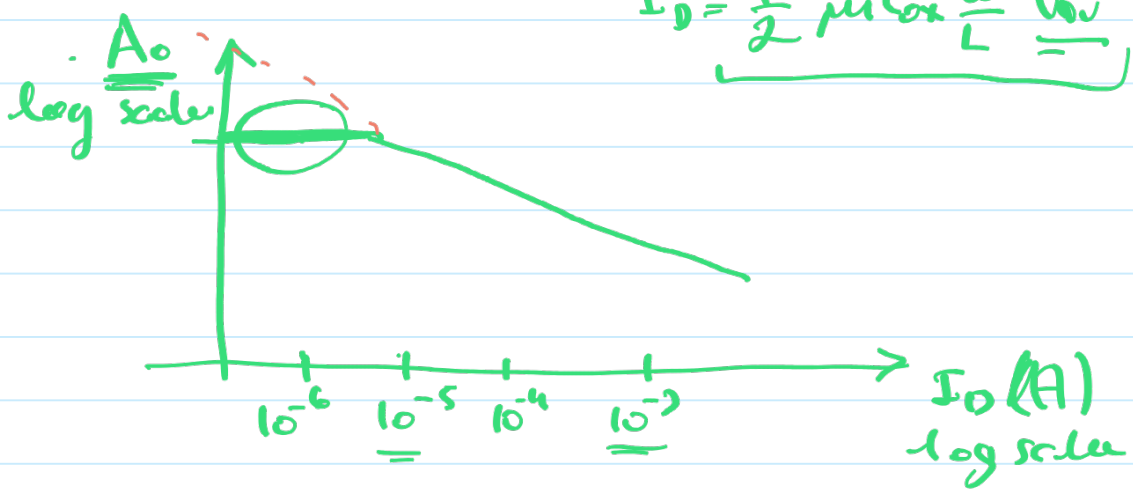
$$r_o = \frac{V_A}{I_D} = \frac{V_A' L}{I_D} = \frac{V_A' L}{I_D}$$

$$A_0 = \frac{I_D}{V_{ov}/2} \cdot \frac{V_A' L}{I_D} = \frac{V_A' L}{V_{ov}/2}$$

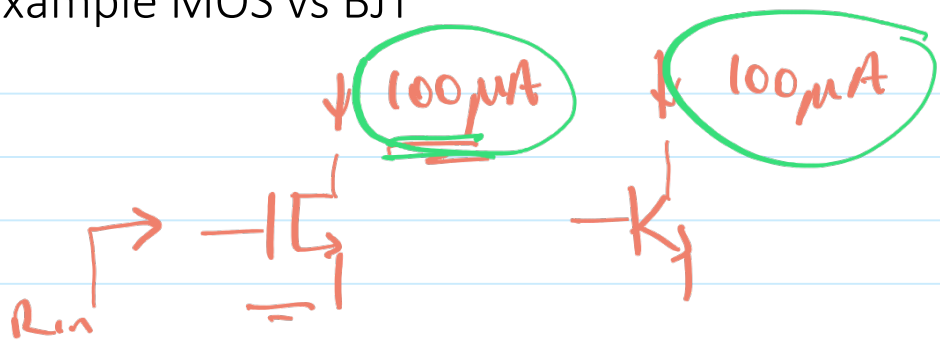
10 – 40

MOSFET

$$I_D = \frac{1}{2}\mu C_{ox} \frac{W}{L} V_{ov}^2$$



Example MOS vs BJT



$L = 0.4\mu m$
$W = 4\mu m$

$\beta = 100$
$V_A = 10$

$\mu_n C_{ox} = 267\ \mu A/V^2$

$V_A' = 10\ V/\mu m$

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{ov}^2$$

$$100\mu = \frac{1}{2}\,267\mu\,10\,V_{ov}^2 \Rightarrow V_{ov} = 0.27V$$

$$g_m = \frac{2I_D}{V_{ov}} = \frac{2\times 100\mu}{0.27} = 0.74\,mA/V$$

$R_{in} = \infty$

$$R_o = r_o = \frac{V_A' L}{I_D} = \frac{10 \cdot 0.4}{100\mu} = 40k\Omega$$

$$A_o = g_m r_o = 0.74m \times 40k = 29.6\ V/V = 30$$

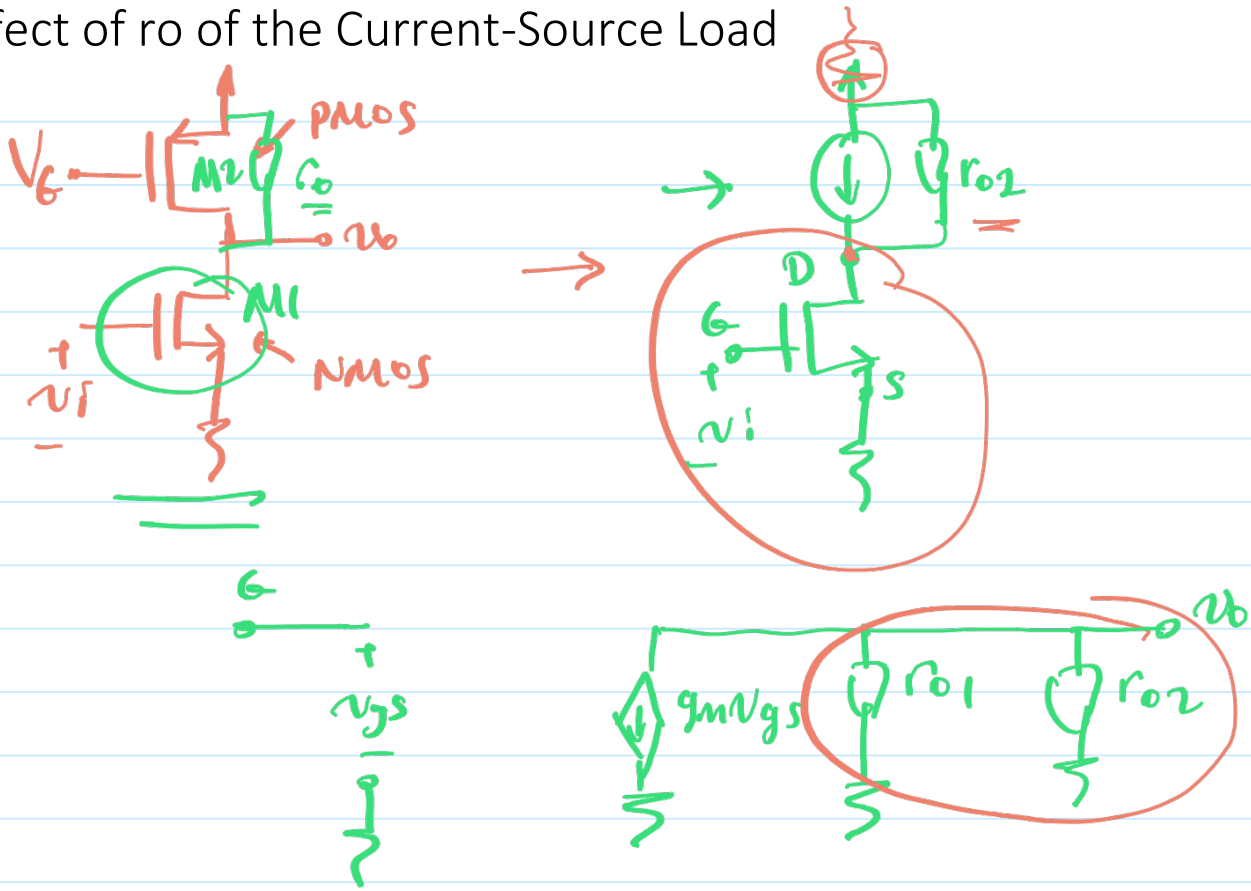
BJT

$$g_m = \frac{I_C}{V_T} = \frac{100\mu}{25m} = 4mA/V$$

$$R_{in} = r_\pi = \frac{\beta}{g_m} = \frac{100}{4m} = 25k \qquad R_o = r_o = \frac{V_A}{I_C} = \frac{10}{0.1} = 100k$$

$$A_o = g_m r_o = 4m \times 100k = 400\ V/V$$

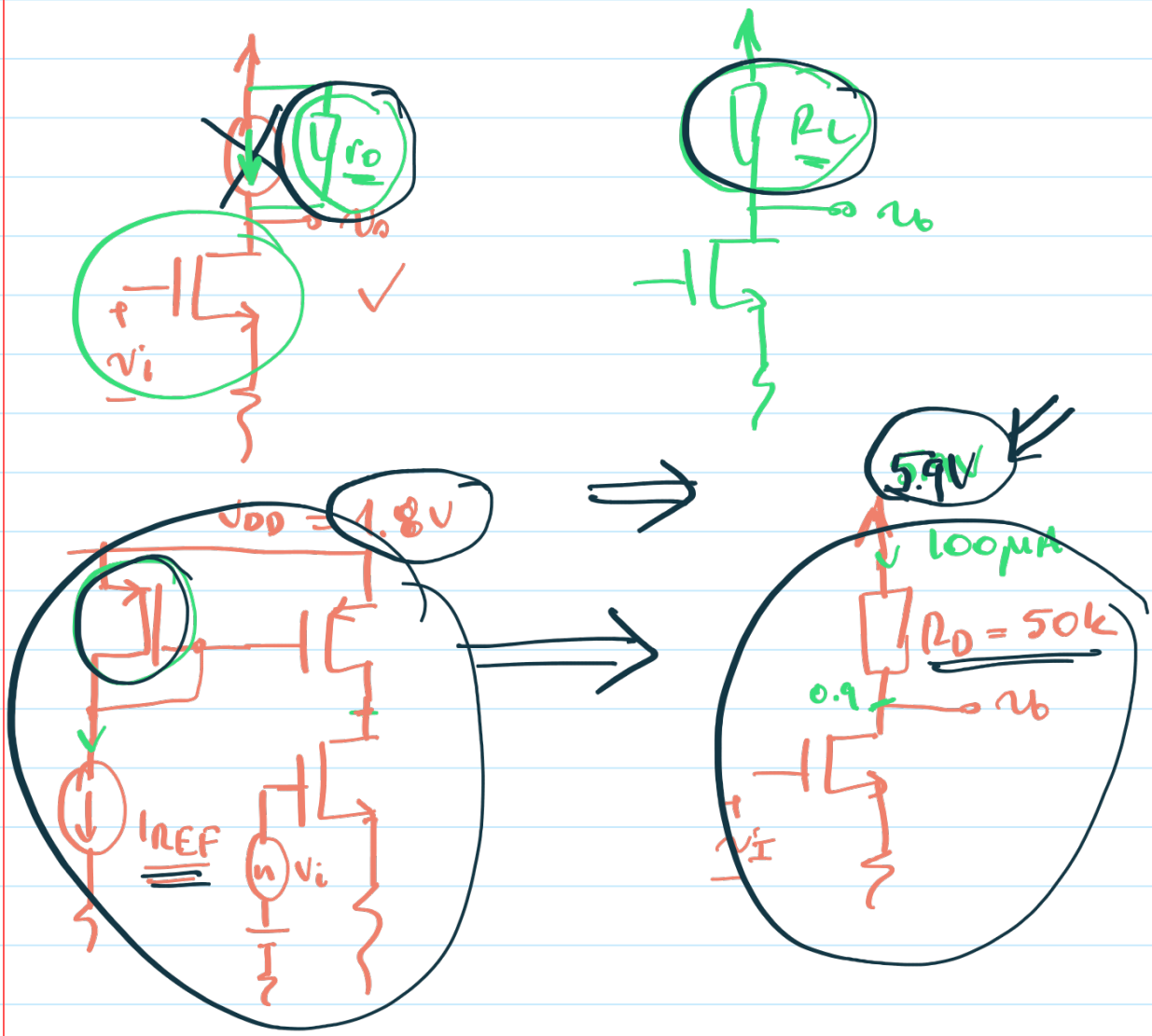
Effect of ro of the Current-Source Load



$$A_v = \frac{v_o}{v_i} = -g_m (r_{o1} \| r_{o2})$$

if $r_{o1} = r_{o2}$ $\quad A_v = \frac{-g_m r_o}{2} = \frac{-A_o}{2}$

Example Current-Source vs Resistor



$|V_{OV}| = 0.2 \Rightarrow V_{OV} = V_{GS} - V_t \Rightarrow V_{GS} = 0.7V$

W/L

$$100\mu = \frac{1}{2} \cdot 100\mu \cdot \frac{W}{L} \cdot 0.04 \Rightarrow \frac{W}{L} = 12.5$$

$$100\mu = \frac{1}{2} \cdot 100 \cdot \frac{W}{L} \cdot 0.04$$

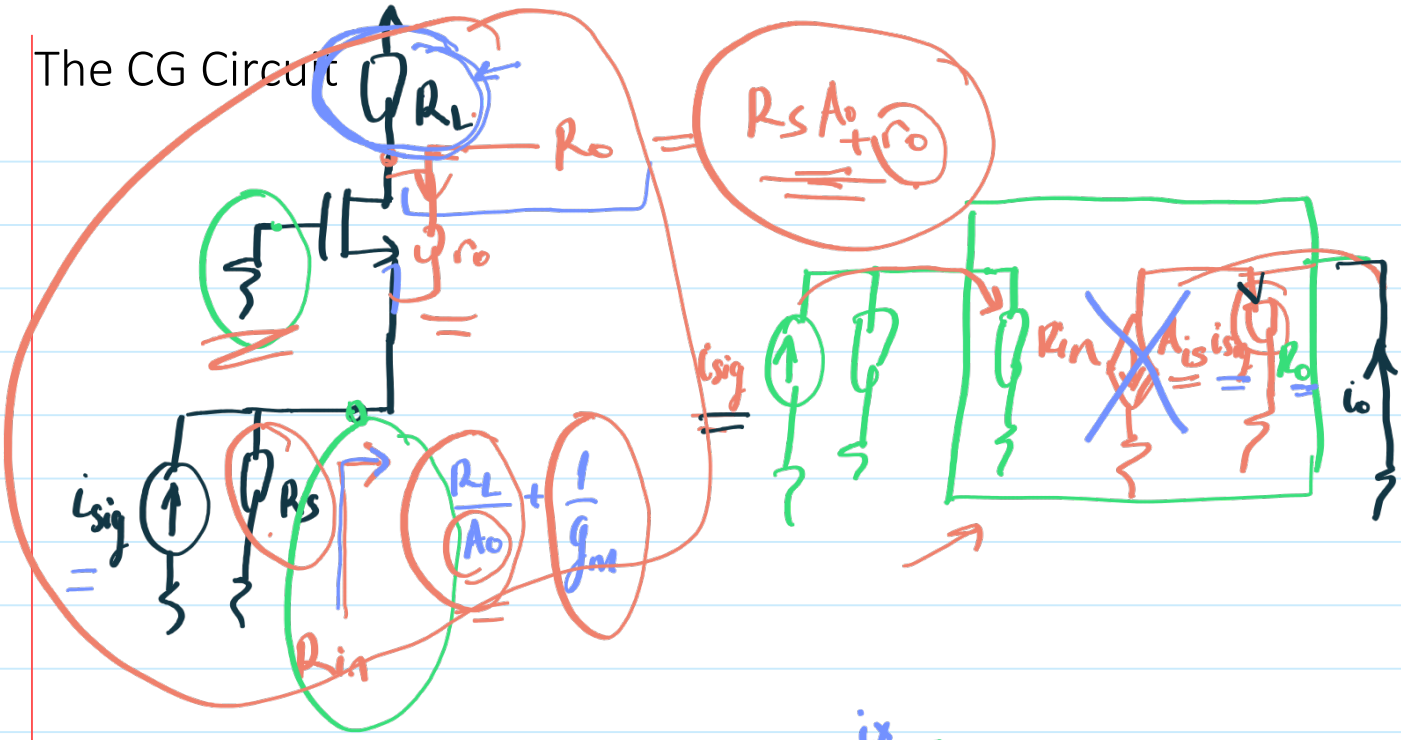
$$\frac{W}{L} = 50$$

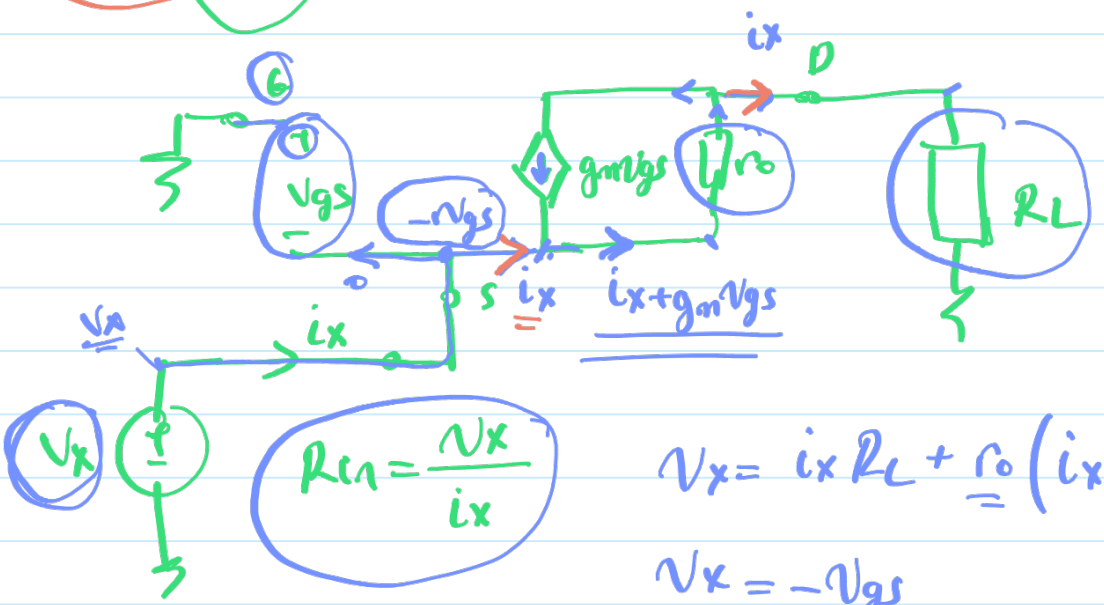
$g_m = 1mA/V \quad r_{o1} = 50k \quad r_{o2} = 50k \quad A_v = -g_m r_{o1} \| r_{o2} = -25\ V/V$

$0.2 < v_o < 1.6$

0.9 ± 0.7 (1.6 / 0.2)

The CG Circuit





$$V_x = i_x R_L + r_o\left(i_x + g_m V_{gs}\right)$$

$$V_x = -V_{gs}$$

$$V_x = i_x R_L + r_o\left(i_x - g_m V_x\right)$$

$$V_x\left(1 + g_m r_o\right) = i_x R_L + i_x r_o = i_x\left(R_L + r_o\right)$$

$$R_{in} = \frac{R_L + r_o}{1 + g_m r_o} \approx \frac{R_L + r_o}{g_m r_o} = \frac{R_L}{g_m r_o} + \frac{1}{g_m}$$

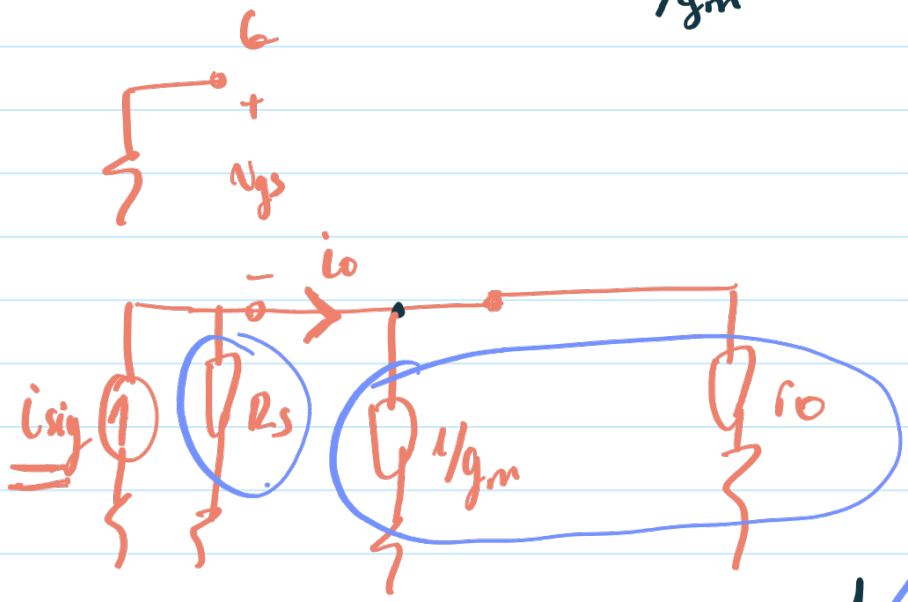
$1 + g_m r_o \approx g_m r_o$

$g_m r_o = A_o$

$\approx 1k$

The CG Circuit

$$\frac{0 + v_{gs}}{1/g_m} = g_m v_{gs}$$



$$A_{is} = \frac{i_o}{i_{sig}}$$

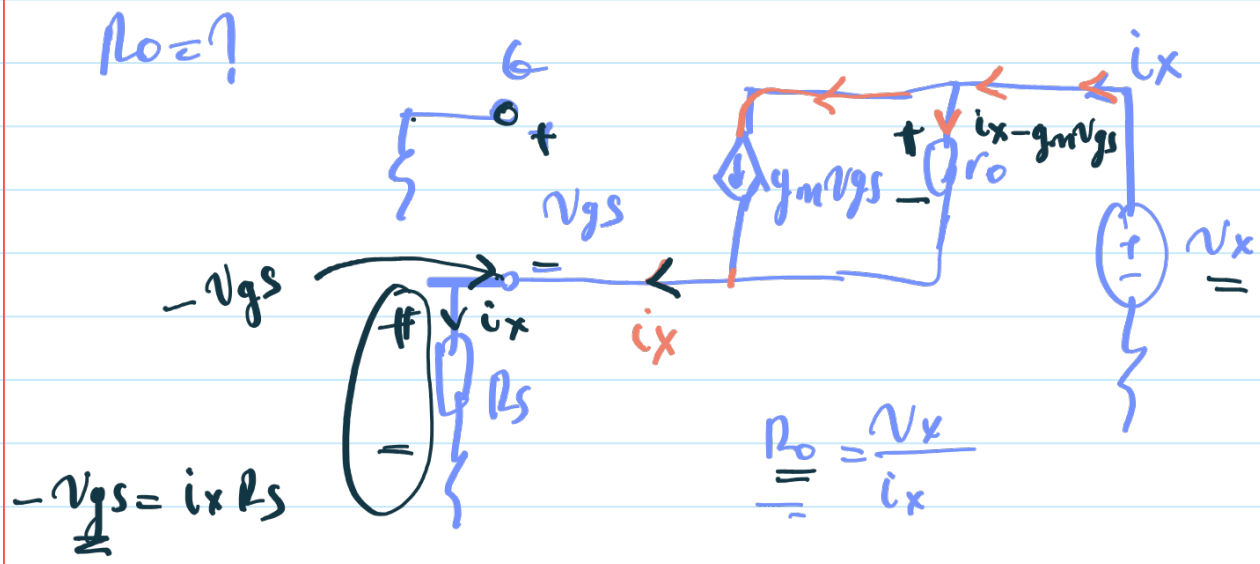
$$A_{is} = \frac{1 + \frac{1}{g_m r_o}}{1 + \frac{1}{g_m r_o} + \frac{1}{g_m R_s}}$$

$n \to 1$

n_0 n_0

$g_m r_o \gg 1$, $g_m R_s \gg 1$

The CG Circuit



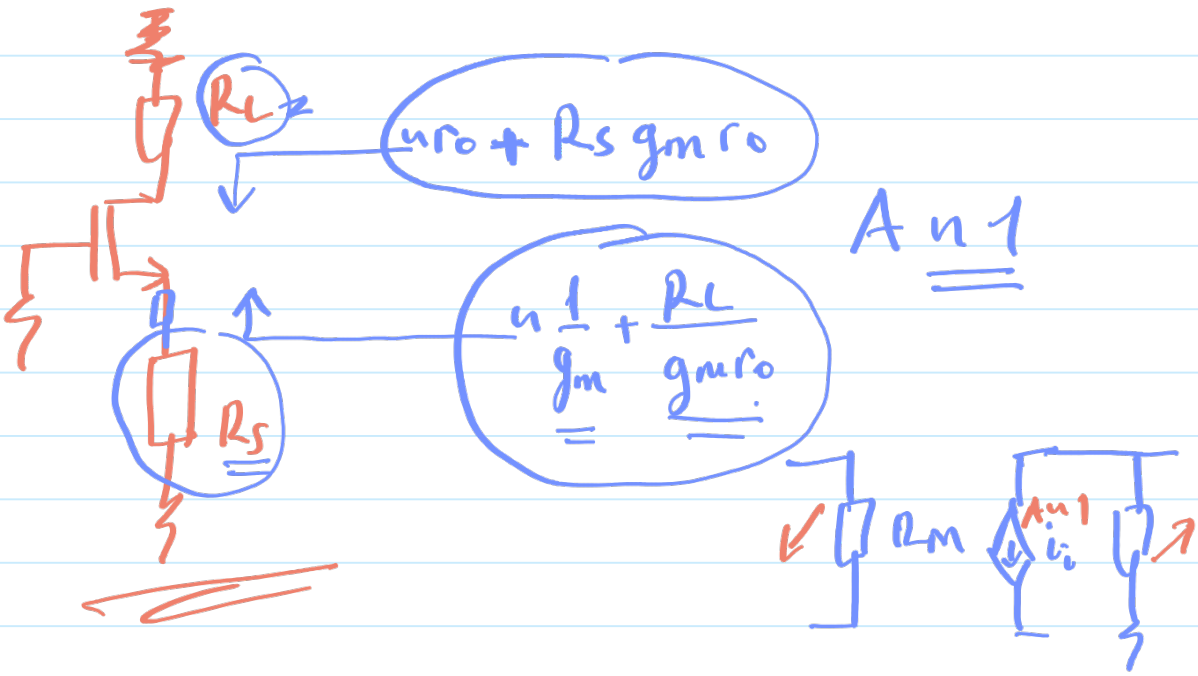
$$r_o\left(i_x + g_m i_x R_s\right) + i_x (R_s) = V_x$$

$$R_o = r_o + R_s\left(1 + g_m r_o\right) = r_o + R_s g_m r_o$$

$$= r_o\left(1 + R_s g_m\right) = r_o + A_o R_s$$

$g_m R_s \gg 1$

$R_o = r_o$



Output Resistance of CS with Rs

The CB Circuit

The CB Circuit

Output Resistance of CE with Re

Template